

Muxin Wei

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RESEARCH INTERESTS

My research interests primarily lie in medical imaging, with a particular focus on multimodal learning, and its application in decision support, in conjunction with clinical reports. The goal is to develop trustworthy and interpretable AI models that improve accuracy and efficiency for real-world healthcare tasks.

EDUCATION

Harbin Institute of Technology

M.Sc., Biomedical Engineering

– Thesis: *Interactive Multimodal Learning for Biomedical Image Segmentation*

Sep 2023 – Mar 2026

Harbin, China

Dongguan University of Technology

B.Sc., Computer Science and Technology

– Thesis: *A Real-time Vehicle Detection and Localization System Based on YOLOV5*

Sep 2018 – Jun 2022

Dongguan, China

PUBLICATIONS

- [Wei, Muxin, et al. ‘Rep-MedSAM: Towards Real-Time and Universal Medical Image Segmentation’. Lecture Notes in Computer Science, 2025, pp. 57–69.](#)

RESEARCH EXPERIENCE

CT Image Generation via Structured Clinical Reports

Aug 2025 – Present

- Evaluated existing generative methods
- Pretrained 2D and 3D VQ-GAN on CT dataset, scaling from 1 to 30K scans
- Bridged the VQ codebook with text prompts via CLIP guidance
- CT image generation with clinical reports [\[demo\]](#)

CCN ’25 Challenge: Brain Response Prediction with Multimodal Movies

Mar 2025 – Jul 2025

- Aligned stimuli of different modalities to fMRI responses
- Evaluated the performance of multiple visual encoding models to predict fMRI brain activity
- Model building and training for cross-modal fMRI prediction with V-JEPA

Medical Image Segmentation with Open-vocabulary

Sep 2024 – Mar 2025

- Explored interpretability ViT in CLIP using Text-Based Decomposition via hook registration
- Improved medical image segmentation with train-free strategies
- Bridged Rep-MedSAM with text prompts by building a Two-way Cross-Attention Transformer to create a text-guided segmentation model

[CVPR ’24 Challenge: Segment Anything In Medical Images On Laptop](#)

Apr 2024 – Jun 2024

- Developed **Rep-MedSAM** via knowledge distillation, achieving superior convergence in **10 epochs** compared to the baseline’s **1000 epochs**
- Optimized the model for **CPU-based multi-object instance segmentation**, reducing inference latency to **0.20s** (2D) and **7.05s** (3D), representing a **2×** and **2.7×** efficiency gain, respectively
- Achieved significant quantitative improvements with an average DSC of **85.90%** and NSD of **87.07%**, surpassing the baseline by margins of **+2.68** and **+4.36**
- Curated **10+** **datasets** with standardized preprocessing pipelines to validate model universality

RELEVANT PROJECT

- Implementation of two-stage post-training (SFT and DPO) on LLaMA2 7B with 100k medical conversations. (LLM project)
- Implementation of Vanilla Policy Gradient, TRPO, PPO, GRPO. (Reinforcement Learning)
- Fine-tuned Qwen2.5-VL 7B on MIMIC dataset with report summaries and grounding. (VLM project)

AWARDS AND SCHOLARSHIPS

- CVPR 2024 Challenge: Segment Anything In Medical Images On Laptop **Winner Finalist Award (#3/102)**
- **2023 - 2024 Second-Class Graduate Scholarship**
- **2024 - 2025 Second-Class Graduate Scholarship**
- **2025 - 2026 Special-Class Graduate Scholarship**

CONFERENCE ACTIVITY

CVPR 2024 Workshop on Foundation Models For Medical Vision

- Presented a summary of proposed method and results analysis for the challenge

SKILLS

Coding: Python (PyTorch, Transformers, OpenCV, ITK, DeepSpeed, vLLM), C/C++, $\text{\LaTeX}$, Bash Scripting

Tools: Git, Docker, Linux, MySQL

MISCELLANEOUS

Interests: Workout, Football, Coffee

Jul 2022 – Jul 2023: A gap year focusing on postgraduate entry examination

Sep 2025: Certified in First Aid training with CPR & AED